

CERTIFICATE OF TEST



Page 01 of 03

Certification Date
9-JAN-2026

CUSTOMER ORDER NUMBER

PO001839

EARLE M. JORGENSEN COMPANY
6201 LUMBERDALE
HOUSTON TX 77092

Invoice Number
T578785

CUSTOMER PART NUMBER

0001

* TAWAIN *

SOLD TO: DOWNHOLE & DESIGN INTL CORP

1443 PROSPECT BOULEVARD
HOUMA LA 70364

SHIP TO:

DOWNHOLE & DESIGN INTL CORP

1443 PROSPECT BOULEVARD
HOUMA LA 70364

Description: 17-4 HR RT BAR - COND DBL H1150 - NACE MR0175
4-1/4 RD X 12' R/L Line Total: 141 IN
HEAT: SR3211-60 ITEM: 510637

Specifications:

ASTM A564 19A

NACE MR0103

AMS-H-6875C

NACE STD MR0175

ISO 17945

API 6D APPENDIX F

ISO 15156-3

API 6A PSL3 18

EN 10204 3.1

CHEMICAL ANALYSIS

C	SI	MN	P	S	NI	CR	MO
0.02	0.44	0.62	0.023	0.001	4.25	15.04	0.34
CU	NBTA	NB	TA				
3.26	0.23	0.23	0.0017				

RCPT: R917177

VENDOR: GLORIA MATERIAL TECHNOLOGY

CTRY OF MELT/POUR: TAIWAN

CTRY OF MANUFACTURE: TAIWAN

MECHANICAL PROPERTIES

	YLD STR	ULT TEN	%ELONG	%RED	HARDNESS
DESCRIPTION	KSI	KSI	IN 02 IN	IN AREA	RC
DBL H1150	118.0	140.0	21.5	61.0	32
	YLD STR	ULT TEN	%ELONG	%RED	HARDNESS
DESCRIPTION				IN AREA	RC
SURFACE					32

The above data were transcribed from the manufacturer's Certificate of Test after verification for completeness and specification requirements of the information on the certificate. All test results remain on file subject to examination.

We hereby certify that the material covered by this report will meet the applicable requirements described herein, including any specification forming a part of the description.

The willful recording of false, fictitious, or fraudulent statements in connection with test results may be punishable as a felony under federal statutes.

Material did not come in contact with mercury while in our possession.

SHANE LU

Manager, Quality Assurance

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GRAIN SIZE :7 -

IMPACT TEST		UOM ft-lbs				% LAT		
TYPE	TEMP	ORNT	SMPL#1	#2	#3	AVG	SHEAR EXPN	DESCRIPTION
CHARPY	-75 F	LONG	68.0	80.0	77.0	75.0		10MMx10MM

MATERIAL IS FREE FROM MERCURY CONTAMINATION
NO WELD REPAIR PERFORMED ON MATERIAL
THERMAL TREATMENT: OK
MACRO: OK
MICRO1: OK

COMMENTS

MELT PRACTICE: EAF/LHF/VOD
REDUCTION RATIO: 9.7:1
TENSILE SPECIMEN SIZE .500"
TEST PIECE IS A PROLONGATION OF THE BAR
HEAT TREATMENT:
SOLUTION ANNEALED @ 1900 DEG F FOR 3.1306 HOURS - AIR COOLED
TO BELOW 90 DEG F
ANNEALED FURNACE MONITORED BY ATMOSPHERIC THERMOCOUPLES
HARDENED @ 1150 DEG F FOR 4.1021 HOURS - AIR COOLED TO BELOW 90 DEG F
HARDENED @ 1150 DEG F FOR 4.1021 HOURS - AIR COOLED TO BELOW 90 DEG F
AGING FURNACE MONITORED BY CONTACT SURFACE THERMOCOUPLES
PASSED UT TEST PER API 6A PSL3
HEAT TREAT LOT NUMBER: 60
LATERAL EXPANSION: 1.15/1.09/1.17 (MM)
SHEAR PERCENTAGE: 76/76/72

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4-1/4 RD X 12' R/L
HEAT: SR3211-60 ITEM: 510637 Line Total: 141 IN
DELTA FERRITE: 0.3%
MATERIAL IS FREE FROM RADIOACTIVITY
YIELD TAKEN AT 0.2% OFFSET

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SHANE LU

A handwritten signature in black ink, appearing to read 'Shane Lu'.

Manager, Quality Assurance



GLORIA MATERIAL TECHNOLOGY CORP. INSPECTION CERTIFICATE



Messrs: EARLE M. JORGENSEN COMPANY

Order No.: 2024005306-2

File No.: 2025001057-A

Grade: 17-4 DBL H1150 510637

Size: 4-1/4"

Heat-Lot No.: SR3211-60

Condition: HF-Solution Annealed-Double Aged-Smooth Turned(-Stress Relief)

P.O.No.: P351180-453

Date: 04/02/2025

Weight: 2161KG 4764LB

P'cs: 9

Chemical Composition (wt%)

	C	Si	Mn	P	S	Ni	Cr
Min.						3.00	15.00
Max.	0.07	1.00	1.00	0.025	0.025	5.00	17.50
Result	0.02	0.44	0.62	0.023	0.001	4.25	15.04
	Mo	Cu	Nb+Ta	Nb	Ta		
Min.		3.00	0.15	5C			
Max.	0.50	5.00	0.45	0.45			
Result	0.34	3.26	0.23	0.23	0.0017		

Mechanical Properties Spec.

	Hardness	Hardness	Hardness	Hardness	Grain Size	Delta-Ferrite			
	1/2R	1/2R(2)	Surface	Surface(2)	ASTM E112	AMS 2315			
Condition					T	L			
Spec.Min.					5				
Spec.Max.	33HRC	33HRC	33HRC	33HRC		5%			
Result	32HRC	32HRC	32HRC	32HRC	7	0.3%			
	Impact(1)	Impact(2)	Impact(3)	Impact(Ave)	LE(1)	LE(2)	LE(3)	LE(Ave)	Shear%(1)
	DB H1150	DB H1150	DB H1150	DB H1150	DB H1150	DB H1150	DB H1150	DB H1150	DB H1150
	1/2R L	1/2R L	1/2R L	1/2R L	1/2R L	1/2R L	1/2R L	1/2R L	1/2R L
Condition	-75°F	-75°F	-75°F	-75°F	-75°F	-75°F	-75°F	-75°F	-75°F
Spec.Min.	20Ft.Lbs	20Ft.Lbs	20Ft.Lbs	20Ft.Lbs					
Spec.Max.									
Result	68Ft.Lbs	80Ft.Lbs	77Ft.Lbs	75Ft.Lbs	1.15mm	1.09mm	1.17mm	1.14mm	76%
	Shear%(2)	Shear%(3)	Shear%(Ave)						
	DB H1150	DB H1150	DB H1150						
	1/2R L	1/2R L	1/2R L						
Condition	-75°F	-75°F	-75°F						
Spec.Min.									
Spec.Max.									
Result	76%	72%	75%						



Peter Liu
Quality Manager
Peter Liu

intertek

Our quality and environment management systems have been certified by AS 9100(QMS), ISO 9001(QMS), IATF 16949(QMS), ISO 17025(Lab), ISO 14001(EMS), ISO 50001(EMS), ISO 45001(OH&S) and NADCAP. We hereby certify that the material described herein has been manufactured and tested with satisfactory results in accordance with the requirement of the above material specification. Inspection Certificate comply with EN 10204 3.1.

台灣台南市新營區新中路35號 / NO.35, HSIN-CHUNG RD., HSIN YING, TAINAN, TAIWAN

TEL: 886-6-6520031 / FAX: 886-6-6230877

http://www.gmtc.com.tw



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Date: 04/02/2025

Weight: 2161KG 4764LB

P'cs: 9

Tensile Test

	Elongation(A)	Tensile Strength(Rm)	Yield Strength(0.2% Rp)	Reduction of Area(Z)
Condition	23.1°C	23.1°C	23.1°C	23.1°C
	DB H1150	DB H1150	DB H1150	DB H1150
	1/2R L	1/2R L	1/2R L	1/2R L
	(Ave)	(Ave)	(Ave)	(Ave)
Unit	%	KSI	KSI	%
Min.	15	135	105	35
Max.				
Result	21.5	140	118	61

Heat Treatment Condition

Condition	Temperature	Time/hour	Treatment
SOLUTION	1900°F	3.1306	FAST AIR COOL TO BELOW 90°F
DOUBLE AGED	1150°F	4.1021	AIR COOL TO BELOW 90°F

ACTUAL REDUCTION RATIO: 9.7:1

COUNTRY OF ORIGIN :GMTC, TAIWAN.

COUNTRY OF MELTING:TAIWAN

Specification:

1. NACE MR 0175/ISO 15156-3(2015).
2. NACE MR 0103/ISO 17945(2015).
3. ASTM A564/A564M-19a.
4. API 6A PSL3-2018.
5. AMS-H-6875C.
6. API 6D APPENDIX F.

Remark:

1. MANUFACTURING PROCESS: AIR MELTING (EAF+LHF+VOD), HOT FORGING OR ROLLING.
2. MATERIAL IS FREE FROM KNOWN CONTACT WITH MERCURY AND RADIUM.
3. MATERIAL IS FREE FROM WELDS OR WELDING REPAIRS.
4. ULTRASONIC TEST CONFORM TO API 6A PSL3.



Intertek

ACCREDITED Heat Treating
Material Testing Laboratory
Nondestructive Testing

Peter Liu
Quality Manager
Peter Liu

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P.O.No.: P351180-453

Date: 04/02/2025

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P'cs: 9

5. MACRO STURCTURE: OK.

6. MICRO STRUCTURE: OK.

7. LONGITUDINAL IMPACT SPECIMEN SIZE IS 10mmX10mmX55mm.

8. LONGITUDINAL TENSILE SPECIMEN SIZE IS D (DIAMETER):0.5 INCH, G (GAGE LENGTH):2 INCH, TEST AT THE ROOM TEMPERATURE.

9. AIR COOLED BELOW 90 DEGREES F AFTER ANN AND AGING CYCLES.

10. AGING CYCLES TO BE MONITORED BY CONTACT SURFACE THERMOCOUPLE.

11. THE TEST PIECE IS A PROLONGATION OF THE BAR.

12. 17-4 DBL H1150 IS THE SAME WITH AISI 630 DBL H1150.

13. THE FURNACE IS MONITORED BY FURNACE ATMOSPHERE.

14. THE HEAT TREATMENT FURNACE IS QUALIFIED ACCORDING TO AMS-H-6875C.

15. MECHANICAL PROPERTIES TESTED PER ASTM A370.

16. THE BATCH TYPE FURNACES ARE USED FOR HEAT TREATMENT IN THE PRODUCTS.

17. FURNACES ARE CALIBRATED IN ACCORDANCE WITH AMS 2750E.

18. THE FURNACE CALIBRATION COMPLIES WITH API 6A 21ST EDITION ANNEX M.



ACCREDITED

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